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PAPER

Boron-based electrolyte solutions with wide electrochemical windows for rechargeable magnesium batteries†

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Recently, we have developed a boron based electrolyte system with outstanding electrochemical performance, formed through the reaction of tri(3,5-dimethylphenyl)borane (Mes_3B) and PhMgCl in THF, for rechargeable magnesium batteries. In this paper, the main components and equilibria of the unique boron based electrolyte solutions are identified by NMR, single-crystal XRD, *etc.* The results prove that the solutions contain various magnesium species, such as $Mg_2Cl_3^+$, $MgCl^+$, Ph_2Mg and the tetracoordinated boron anion $[Mes_3BPh]^-$. Fluorescence spectra and Raman spectroscopy analyses indicate that the high anodic stability (*ca.* 3.5 V *vs.* Mg reference electrode (RE)) of the Mes_3B–$(PhMgCl)_2$ electrolyte is attributed to the non-covalent interactions between the anion $[Mes_3BPh]^-$ and Ph_2Mg. Furthermore, the air sensitivity of the boron based electrolyte and its electrochemical stability on the different current collectors are studied. Finally, the reversible electrochemical process of Mg intercalation into a Mo_6S_8 cathode confirms that the boron based electrolyte could be practically used in rechargeable Mg battery systems.

1. Introduction

Electronic vehicles (EVs) may present one of the best solutions for the transportation sector to reduce our dependence on petroleum as well as emissions of environmental pollution. Cost-effective and abuse-tolerant rechargeable batteries with high energy densities and long cycle life are key to the growth of EVs.[1] Lithium-ion batteries have become a hot research topic due to their high energy densities, long cycle life and environmental friendliness. However, their use in EVs is still handicapped by safety issues and high costs.[2,3] Therefore, the development of new types of green and safer, less-expensive rechargeable battery systems is critical and essential.

Recently, rechargeable Mg batteries have gained more and more attention as next-generation energy storage devices because of their potential advantages (safety, low cost and long life) in large-scale applications.[4,5] However, the development of rechargeable Mg batteries suffers from the fatal problem that the reversible cycling of the Mg negative electrode seems to be quite difficult in most nonaqueous organic electrolytes due to its passivating characteristics.[4] So it's very important to find suitable

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Broader context

The demand for green secondary batteries is continuously increasing because of sustainable energy and environmental issues. This has urged scientists to search for new and advanced battery systems to meet the demands. Currently, rechargeable Mg batteries have gained more and more attention as next-generation energy storage devices because they have potential advantages in heavy load applications. However, the lack of suitable electrolyte solutions with wide electrochemical windows is a bottleneck in the development of high-energy rechargeable Mg batteries. Here we report a high-performance electrolyte system based on organic boron Mg complex salts, which presents the highest anodic potential up to 3.5 V *vs.* Mg RE. The main components and equilibria of the unique boron based electrolyte solutions are identified by NMR, single-crystal XRD, *etc.* The correlation between the electrochemical anodic stability and the identified solution equilibrium species is investigated. Furthermore, this boron based electrolyte system has recovered ability after partially losing its performance after being exposed to air and is less corrosive. Good compatibility with the Mo_6S_8 Chevrel phase cathode confirms that the boron based electrolyte could be practically used in rechargeable Mg battery systems. These outstanding performances will promote the development of high-energy rechargeable Mg batteries efficiently.

electrolyte systems for developing high-energy rechargeable Mg batteries, in which reversible Mg deposition, high ion conductivity and wide electrochemical windows can be obtained.

It is well known that the Mg electrode displays a highly reversible behavior in organometallic Mg salt solutions such as Grignard salts in ethereal solutions (RMgX, R = alkyl, aryl groups; X = Cl, Br).[6–8] However, they cannot be used in Mg batteries because of their narrow electrochemical windows.[9] A breakthrough was made by Aurbach *et al.*, who developed a family of electrolyte solutions based on organohaloaluminate Mg salts $Mg(AlCl_{4-n}R_{n'}R'_{n''})_2$ (R, R′ = alkyl or aryl groups, $n'+n'' = n$) in ethereal solvents. The optimized composition of the THF solution of the $Mg(AlCl_2BuEt)_2$ complex[10] and $AlCl_3$–$(PhMgCl)_2$ complex[11] exhibited an improved anti-oxidation capability and excellent reversible Mg deposition–dissolution characteristics, which greatly promoted the development of rechargeable Mg batteries.[5] In the year 2011, Kim *et al.* reported that the crystal of $[Mg_2Cl_3$–6THF][HDMSAlCl$_3$], which was generated from the electrolyte of the Grignard hexamethyldisilazide magnesium chloride (HMDSMgCl) and $AlCl_3$ in a 3 : 1 ratio, can significantly enhance the oxidation potential up to 3.2 V (*vs.* Mg RE) as it was used as a solute for the electrolyte.[12]

Organic boron Mg complex salts (OBMCs) would be also promising electrolyte candidates for rechargeable Mg batteries. In 1990, Gregory *et al.*[4] reported the first boron-based electrolyte of $Mg(BPh_2Bu_2)_2$ solution (Ph and Bu represent phenyl and butyl groups, respectively) which showed an efficient reversible Mg deposition–dissolution process, and an improved anodic stability compared with some common Grignard salt solutions (1.9 V *vs.* 1.5 V). Aurbach *et al.* also investigated THF solutions of Bu_2Mg as Lewis bases, together with different boron based Lewis acids (BPh_2Cl, $BPhCl_2$, $B[(CH_3)_2N]$, BPh_3, BEt_3, BBr_3, BCl_3 and BF_3). However, the results showed that these boron based electrolytes have low Mg cycling efficiencies and poor anodic stabilities (about 1.5 V).[10] Apart from these publications, we are not aware of any published research on boron based electrolytes for rechargeable Mg batteries. Recently, we have developed a family of novel boron based electrolyte solutions with high ionic conductivity, excellent Mg deposition reversibility (*ca.* 100%) as well as the highest anodic potential to date (3.5 V *vs.* Mg RE).[13] However, a complete understanding of the structure–function relationships of this boron based electrolyte remains elusive. In the present work, the identification of the equilibrium species in the boron based electrolyte solutions is implemented by NMR, single-crystal XRD analyses, Raman spectroscopy and fluorescence spectra, which help to further understand the structure–function relationships of the novel boron based electrolyte system. Furthermore, the air sensitivity of the boron based electrolyte and the electrochemical stability of the different current collectors in the boron based electrolyte are studied. Lastly, we investigate the charge–discharge performance of a Mg–Mo_6S_8 battery containing the boron based electrolyte solution.

2. Experimental section

2.1 Synthesis of Mes_3B

The synthetic work was conducted under strict inert gas conditions in a 3-neck flask equipped with a N_2 inlet, dropping funnel, and gas outlet to a liquid paraffin bubbler using a modified organometallic route involving the direct reaction of organic halide, Mg, and $BF_3\cdot OEt_2$. All glass wares were acid-washed and vacuum dried prior to use. Mg turnings (industrial-grade, 0.528 g, 22 mmol) were put into the reaction flask, a crystal of iodine and $BF_3\cdot OEt_2$ (46%, 2.035 g) were introduced to the reaction flask while maintaining the nitrogen atmosphere, and the reaction was initiated by the dropwise addition of 1-bromo-3,5-dimethylbenzene (3.70 g 20 mmol) in anhydrous ether (20 ml). Next, the remaining portion was added slowly such that the ether refluxed gently. The reaction solution was stirred for 8 h at room temperature, after which, the solid residue was separated and the solvent Et_2O was removed under reduced pressure. The resulting solid powder was dried under vacuum (yield, 2.1 g, 97%). ^{1}H NMR ($CDCl_3$, 400 MHz): δ 2.37 (s, 18H, –CH_3), 7.2–7.26 (m, 9H, –C_6H_3); ^{13}C NMR ($CDCl_3$, 100 MHz): δ 25.8 (–CH_3), 132.6, 136.0, 136.4 (Ph). ^{11}B NMR ($CDCl_3$, 128.4 MHz): δ −20.23.

2.2 Electrolyte solution synthesis

All the experiments were conducted using standard Schlenk techniques under dry nitrogen, and the THF was thoroughly dried by sodium and benzophenone. The boron based organic Mg electrolytes were synthesized by the reaction of Mes_3B, Lewis acid and Grignard reagent Lewis bases in different concentrations and molar ratios following the synthesis procedure of $[Mes_3BPh]^-[MgCl]^+$: firstly, Mes_3B (0.332 g, 1 mmol) was dissolved in highly pure THF (3 ml), and then, PhMgCl–THF solution (Aldrich, 2 M, 0.5 ml, 1 mmol) was slowly added to the Mes_3B solution. The resulting solution was stirred for several hours at room temperature. The solvent was removed under reduced pressure, and the product was afforded as a solid powder in essentially quantitative yield. ^{1}H NMR (d_8-THF, 400 MHz): δ 2.04 (s, 18H, methyl protons), 6.31 (s, 3H, aromatic protons), 6.61–6.89, 7.20–7.25 (m, 11H, aromatic protons); ^{13}C NMR (d_8-THF, 100 MHz): δ 23.4 (methyl carbons), 122.5, 122.9, 124.6, 126.8, 130.1, 134.0, 136.5, 138.2 (aromatic carbons); ^{11}B NMR ($CDCl_3$, 128.4 MHz): δ −26.2. The second generation of the Al-based electrolyte was synthesized according to the literature.[11]

2.3 Electrochemistry

Cyclic voltammograms (CVs) were measured with a three-electrode arrangement using a CHI604A electrochemical workstation (Shanghai, China). The working electrodes were different metal disc electrodes (Pt, Cu, Al, Ni, SS = stainless steel, area: 0.0314 cm^2) and magnesium ribbons served as counter electrode and reference electrode (RE). All of the electrodes were polished with a corundum suspension and rinsed with dry acetone before use. The Mo_6S_8 cathode material was synthesized by a molten salt method according to the literature,[14] the Mo_6S_8 composite electrode was prepared by casting and pressing a 8 : 1 : 1 weight-ratio mixture of Mo_6S_8, super-P carbon powder and polyvinylidene fluoride (PVDF) binder dissolved in *N*-methyl-2-pyrrolidinone (NMP) onto a Ni current collector followed by drying in a vacuum at 120 °C. Rechargeable Mg batteries were fabricated in standard 2016 coin cells using a thin Mg disc anode, Entek PE separator, boron based electrolyte and Mo_6S_8

composite cathode. The charge–discharge measurements of the coin cells were carried out on a land battery measurement system (Wuhan, China) with a cutoff voltage of 1.7/0.5 V *vs.* Mg RE.

2.4 Spectral studies

NMR spectra were recorded using a Bruker Avance-III-400 spectrometer. Raman spectra were measured with a Senterra R200-L (Bruker Optics) using a He–Ne excitation laser (532 nm) in a sealed capillary. All the experiments were performed at room temperature (25 $\pm$ 2 °C). The fluorescence spectra were recorded on a Hitachi F-4500 spectrofluorometer.

2.5 Crystallography

A single crystal *ca.* $0.20 \times 0.15 \times 0.14$ mm in size was selected for data collection. The crystal was coated with inert crystallography oil, and rapidly transferred (slightly wet) to the cold N_2 stream on the instrument. The measurements were carried out using the Bruker "Kappa Apex II" system. The structure was solved by direct methods and expanded routinely. The model was refined by the full-matrix least-squares analysis of F_2 against all reflections. All non-hydrogen atoms were refined with anisotropic thermal displacement parameters. Unless otherwise described, hydrogen atoms were included in calculated positions. Structure calculation and refinement were performed with the SHELXS-97 program.

Crystal data for [Mg_2Cl_3–6THF][Mes_3BPh]: Mr = 495.48; monoclinic, space group *P2(1)/n*, $a = 16.076(4)$ Å; $b = 23.952(5)$ Å; $c = 17.490(4)$ Å, $\alpha = 90°$; $\beta = 108.075(3)°$; $\gamma = 90°$, $V = 6403(2)$ Å, $Z = 4$, $\rho_{calcd} = 1.028$ mg m^{-3} $R_1 = 0.1031$ $wR_2 = 0.2969$.

3. Results and discussion

In our present strategy, a series of new structurally well-defined Lewis acids of triarylborates (BR_3) are synthesized using a modified organometallic route method,[15] and novel boron based electrolyte solutions have been developed using the BR_3 (R = methyl- or fluoro-substituted phenyl groups) Lewis acid complex with Grignard salts (RMgX, R = aryl, alkyl *etc.* and X = Cl, Br) as Lewis bases at different acid–base ratios. We examine the boron based electrolyte solutions in terms of the possibility of obtaining reversible Mg deposition and higher anodic stability of the electrolytes using a cyclic voltammetric method. It is found that the electrochemical properties of the boron based electrolytes is greatly influenced by several factors including the type of Lewis acid, Lewis base and acid–base ratio (see Fig. S1†). A highly important result is that methyl-substituted aryl borates show better electrochemical performance than fluoro-substituted ones as complexes toward PhMgCl Grignard reagents. However, a rigorous study of the reason behind this is beyond the scope of this work.

Among these BR_3, tri(3,5-dimethylphenyl)borane (Mes_3B, Mes = 3,5-$Me_2C_6H_3$, see Fig. S2†) as a Lewis acid complex with the PhMgCl Grignard reagent shows the best electrochemical performance, which is highly promising for use in rechargeable magnesium batteries. Therefore, the following work is concentrated on the study of Mes_3B–$(PhMgCl)_x$ complex solutions.

Fig. 1 shows the typical cyclic voltammetry (CV) curve of a Pt working electrode in Mes_3B–$(PhMgCl)_2$/THF solution. Two distinct characteristics have been observed. First, the electrochemical deposition–dissolution of Mg in the electrolyte is fully reversible at almost 100% efficiency (calculated according to the ratio of the electric charge amount of Mg dissolution to that of Mg deposition), which has been further confirmed by chronopotentiograms of Mg deposition and dissolution. Second, the electrolyte exhibits a wide electrochemical window until *ca.* 3.5 V (*vs.* Mg RE), which is one volt more positive than that of the Mes_3B–PhMgCl solution and may be the widest of that reported in the literature to date.[11] It was assumed that the most vulnerable oxidative species in the electrolyte governs the solutions anodic stability.[16] Reasonably, the Mes_3B–$(PhMgCl)_2$ electrolyte is a mixed solution of $[Mes_3BPh]^-[MgCl]^+$ compounds and another one equivalent of PhMgCl, and its anodic stability should be very low if determined by PhMgCl. However, it is surprising that the Mes_3B–$(PhMgCl)_2$ electrolyte shows the highest anodic stability. We hypothesized that this unexpected high electrochemical performance of the Mes_3B–$(PhMgCl)_2$ electrolyte solution might be as a result of the formation of distinct products between $[Mes_3BPh]^-[MgCl]^+$ and PhMgCl.[13]

To confirm our speculations, we performed a multinuclear NMR analysis for elucidating the structural information of the species in the Mes_3B–PhMgCl and Mes_3B–$(PhMgCl)_2$ electrolytes. Structural information can be obtained by running spectra for each of the nuclei: ^1H, ^{13}C and ^{11}B. The ^{11}B NMR spectrum of Mes_3B–PhMgCl (Fig. 2b) shows a peak at $\delta = -26.2$ ppm, which is typically ascribed to a tetra-coordinated boron anion.[17–19] Here the binding of the phenyl moiety to boron is confirmed. Combined with the ^1H and ^{13}C NMR spectra analyses (Fig. S2 and S3†), a stable $[Mes_3BPh]^-[MgCl]^+$ compound, which was formed by the transmetallation reaction of Mes_3B with PhMgCl compound in a 1 : 1 ratio, can be structurally well-defined. Notably, the ^{11}B NMR spectrum of Mes_3B–$(PhMgCl)_2$ (Fig. 2c) reveals a high-intensity peak at the same chemical shift as that of Mes_3B–(PhMgCl) ($\delta = -26.2$ ppm), which confirms that the tetra-coordinated boron anion $[Mes_3BPh]^-$ is stable in the electrolyte solution. The ^{13}C NMR

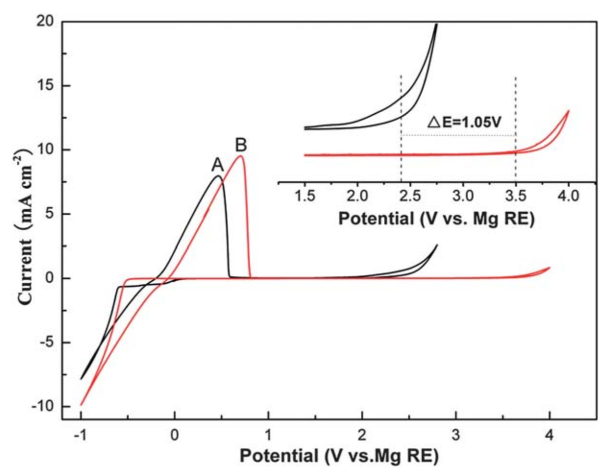


Fig. 1 Typical steady-state voltammograms of Pt electrodes in different electrolytes: 0.4 M Mes_3B–PhMgCl (A) and 0.5 M Mes_3B–$(PhMgCl)_2$ (B), the inset is an enlargement of the 1.5–4 V region.

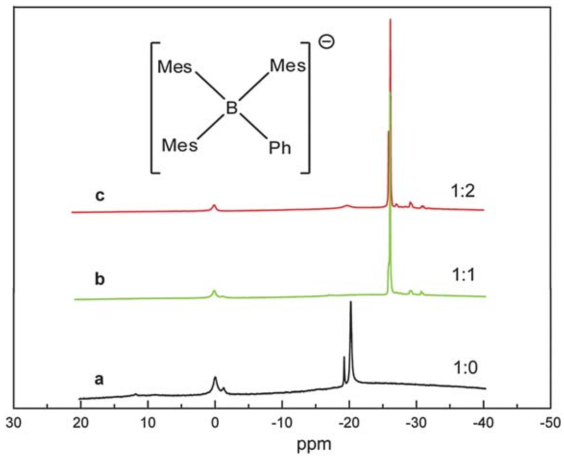


Fig. 2 [11]B NMR spectra measured with boron based electrolyte solutions of the following solutions: (a) Mes_3B/THF, (b) Mes_3B–PhMgCl/THF, (c) Mes_3B–$(PhMgCl)_2$/THF.

analyses (Fig. S3†) show that Mes_3B–$(PhMgCl)_2$ is not a simple mixture of Mes_3B–PhMgCl and PhMgCl in THF solution, suggesting that new equilibria or interactions might be established.

In order to make a deeper insight into the Mes_3B–$(PhMgCl)_2$ electrolyte system, crystallographic analysis has been performed. Upon slowly diffusing hexane into a solution of Mes_3B–$(PhMgCl)_2$ in THF, a large amount of colorless block-like crystals were formed within one or two days, which was determined by single-crystal X-ray diffraction. After selection of a suitable single crystal and subsequent data collection, the structure was determined to be $[Mg_2Cl_3–6THF]^+[Mes_3BPh]^-$ in which the asymmetric unit consists of an ion pair (see Fig. 3). The anion is a boron atom tetrahedrally coordinated by three 3,5-dimethylphenyl groups and one phenyl group. The distances of the B–C bonds (1.65 Å) and the angle of C–B–C (110°) are in the typical range.[20,21] In the counter cation part, each Mg center, coordinated by three THF molecules through the oxygen atoms, is bridged by three chlorine atoms to form a typical two-core structure. The pertinent crystal data and the characteristics of the data collection and refinement are listed in the experimental section.

The electrochemical behavior of the above obtained $[Mg_2Cl_3–6THF]^+[Mes_3BPh]^-$ crystal re-dissolved in THF exhibited a reversible Mg deposition–dissolution process, and an oxidation potential (*ca.* 2.5 V) (see Fig. 4) significantly lower by about 1.0 V than that of the *in situ* Mes_3B–$(PhMgCl)_2$ electrolyte solution, and similar to the value for the Mes_3B–(PhMgCl) electrolyte. This is completely different from several previously reported Al-based electrolyte systems. One structural analogue $[Mg_2Cl_3–6THF]^+[HDMSAlCl_3]^-$ component crystallized *in situ* from a 3 : 1 mixture of HDMSMgCl to $AlCl_3$, showed dramatic improvements in electrochemical stability and coulombic efficiency.[12] Another structural analogue $[Mg_2Cl_3–6THF][EtAlCl_3]$ crystal obtained from the first generation electrolyte of $Mg(AlCl_2BuEt)_2$/THF solution, was electrochemically inactive.[10] The $[Mg_2(\mu\text{-}Cl)_3\cdot 6THF]^+$ found in these three systems, was typically regarded as one of the electroactive species, and responsible for the electrochemical behavior of the electrolyte combined with the other species. The strong influence of the variation of the counter anion on electrolyte properties has also been indicated in the above description. Obviously the interaction among the various species plays an important role in the electrolyte properties, and thus the additional component would generate some unexpected effects on the electrolyte systems. Reasonably, in our case, the high electrochemical performance of the Mes_3B–$(PhMgCl)_2$ electrolyte is governed by the intermediate product formed through weak interactions between $[Mes_3BPh]^-[MgCl]^+$ and PhMgCl. Unfortunately, the exact intermediate product cannot be revealed by crystallographic analysis.

It was noted that significant color changes occurred as the Lewis acid Mes_3B mixed with PhMgCl in different molar ratios, but such phenomena could not be found in the case of mixing Mes_3B with an alkyl Grignard such as EtMgCl. In this respect, the effect of the aromatic moiety on the electrolytes has been examined using fluorescence spectroscopy analysis for different ratios of Mes_3B mixed with PhMgCl. Surprisingly, as shown in Fig. 5, the maximum peak is extensively declined for Mes_3B–$(PhMgCl)_2$ in comparison with Mes_3B–PhMgCl. Also, the solution color was deepened to a brown yellow (see the inset of Fig. 5). The remarkably different optical spectral profiles between the presence and absence of excess amounts of PhMgCl indicate the occurrence of some weak interactions in the case of

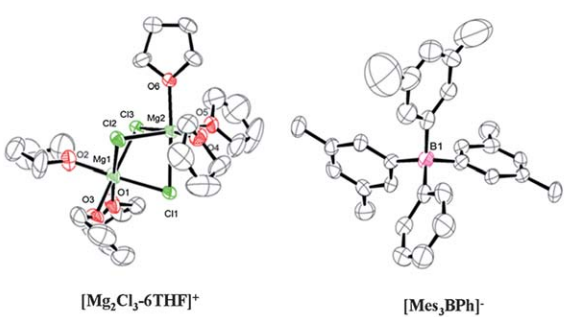


Fig. 3 ORTEP drawing of the molecular structure of the crystallized Mes_3B–$(PhMgCl)_2$ complex electrolyte. Hydrogen atoms have been omitted for clarity.

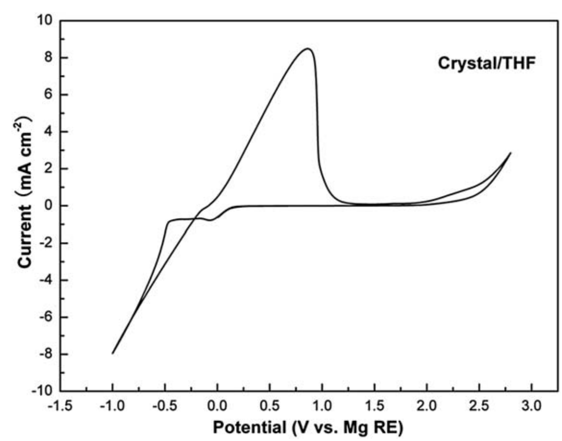


Fig. 4 The steady-state cyclic voltammogram of the Pt electrode in 0.5 M solution of the crystal $[Mg_2Cl_3–6THF]^+[Mes_3BPh]^-$ in THF solvent at a scanning rate of 50 mV S^{-1}.

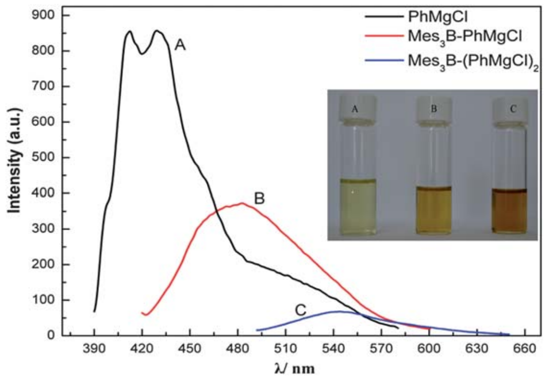


Fig. 5 Fluorescence spectra of 0.5 M solutions of (A) PhMgCl, (B) Mes_3B–PhMgCl, (C) Mes_3B–$(PhMgCl)_2$, in THF. (The colors of the corresponding solutions are shown in the inset.)

Mes_3B–$(PhMgCl)_2$, *e.g.* aggregates[22] or π–π interactions.[23] Furthermore, Raman spectroscopy (see Fig. S4†) presented two unique peaks at 635 and 991 cm^{-1}, typically attributed to the Ph–Mg vibration from the separate PhMgCl in THF.[24] However, the remarkable changes in the scattering peaks in the corresponding regions for the Mes_3B–$(PhMgCl)_2$ electrolyte are likely due to the formation certain kinds of perplexing species through weak interactions. Such results are in line with the analysis of the fluorescence spectra. Although the exact intermediate product is still unclear, we may safely draw the conclusion that the two equivalents of PhMgCl in Mes_3B–$(PhMgCl)_2$ not only provide one equivalent of Lewis base to form the stable cation $[Mes_3BPh]^-$, but also act as critical species to construct effective compositions through weak interactions for improving the electrochemical performance of the whole electrolyte system.

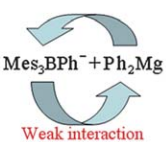
Based on the NMR, single-crystal XRD, Raman and fluorescence spectra analyses, it is possible to elaborate on the reactions and equilibria that take place in each solution (Scheme 1). Eqn (1)–(4) show the reaction paths and the final composition of the Mes_3B–$(PhMgCl)_2$ electrolyte solution. In eqn (1), the stable $[Mes_3BPh]^-$ was generated by transmetallation, in which 1.0 eq. Mes_3B was stoichiometrically consumed by reacting with 1.0 eq. PhMgCl. Eqn (2) presented a typical Schlenk equilibrium from another 1.0 eq. PhMgCl.[25] As a result of eqn (1) and (2), chlorine atom-bridged $Mg_2Cl_3^-$ was formed by coupling $MgCl^+$ with $MgCl_2$ in eqn (3). On the basis of the above three equations, a general eqn (4) of Mes_3B–$(PhMgCl)_2$ in THF was obtained, reflecting the first approximation of the reaction paths and the product distribution for a boron-based electrolyte system. The solution comprises $MgCl^+$, $Mg_2Cl_3^+$, Mes_3BPh^-, as the main active species, all the above Mg–Cl moieties are stabilized by the THF molecules to construct important elements in the electrolyte system.[26] The high anodic stability of the Mes_3B–$(PhMgCl)_2$ electrolyte solution is rationally attributed to the weak interactions between the anion $[Mes_3BPh]^-$ and Ph_2Mg.

$$Mes_3B + PhMgCl \rightarrow MgCl^+ + Mes_3BPh^- \quad \text{Transmetallation} \quad (1)$$

$$2\,PhMgCl \rightleftharpoons Ph_2Mg + MgCl_2 \quad \text{Schlenk equilibrium} \quad (2)$$

$$MgCl_2 + MgCl^+ \rightleftharpoons Mg_2Cl_3^+ \quad (3)$$

$$2\,Mes_3B + 4\,PhMgCl \rightleftharpoons MgCl^+ + Mg_2Cl_3^+ + 2\,Mes_3BPh^- + Ph_2Mg \quad (4)$$



Scheme 1

Air stability is very important for magnesium ion electrolytes. The reported Al-based electrolytes are sensitive to air/moisture. For example, the second generation electrolyte [$(PhMgCl)_2$–$AlCl_3$/THF solution] developed by Aurbach[11] is unable to endure air and no reversible Mg deposition–dissolution processes can be observed after exposing to air for several hours (inset of Fig. 6). As the Mes_3B–$(PhMgCl)_2$ electrolyte solution was tested after exposing it to air for the same amount of time, electrochemical Mg deposition and dissolution was maintained on the whole. However, a significant decline of the electrochemical oxidation potential was observed. Interestingly, it is found that the high oxidation potential can be efficiently recovered again by the addition of another 1.0 eq. of PhMgCl into the electrolyte. In addition, the reversible current peaks for Mg deposition and dissolution are also increased (see Fig. 6). This result clearly indicates the key role of excess amounts of Grignard reagent in our electrolyte system. In particular, such an ability of the Grignard reagent PhMgCl to refresh the high electrochemical performance of the boron-based electrolyte might provide a simple way to overcome the potential shortcomings of the electrolyte system sensitive to air/moisture, facilitating electrolyte processing in practical applications.

The current collector is a key component in rechargeable battery systems. Nowadays, the most common current collectors (Cu, Al, Ni *etc.*) used in lithium ion batteries have poor resistance to corrosion in many organohaloaluminate electrolyte systems. Inert noble metals such as platinum (Pt) have been used to demonstrate the electrochemical performance of organohaloaluminate electrolyte systems. However, they are too expensive to use as current collectors in commercially viable rechargeable Mg batteries. For example, the second generation electrolyte, which was developed by Aurbach,[11] displays an electrochemical stability window of up to 3 V *vs.* Mg RE on the noble Pt working electrode. Nevertheless, the electrochemical

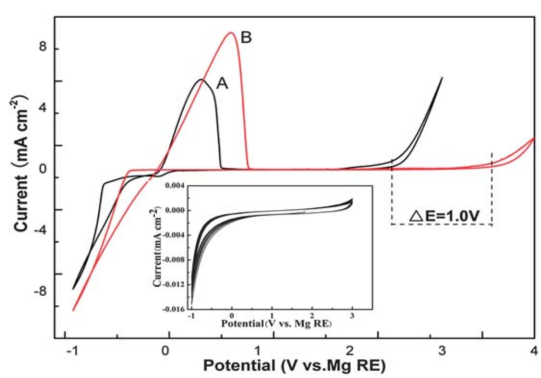


Fig. 6 Steady-state cyclic voltammograms of the Pt electrode in Mes_3B–$(PhMgCl)_2$/THF at a scanning rate of 50 mV S^{-1}: (A) exposed to air for 3 hours, (B) refreshed by adding another 1.0 eq. of PhMgCl. (The inset is the cyclic voltammogram of the Pt electrode in $(PhMgCl)_2$–$AlCl_3$/THF solution exposed to air for 3 hours.)

window is considerably narrow when using non-noble metal current collector materials. As shown in Fig. 7a, Cu, Al, Ni and stainless steel (SS) display stable windows of 1.9, 1.2, 2.1 and 2.0 V (*vs.* Mg RE), respectively, presumably because these metal current collectors are easily corroded in the current magnesium organohaloaluminate electrolytes due to the presence of halides in the anion components of the electrolyte.[27] The oxidative stability of our boron based electrolyte system on these metal electrodes was investigated. The anodic stability of the boron based electrolyte on Cu and SS are 2.0 and 2.2 V, slightly higher than that of the second generation electrolyte. This result indicates that stainless steel coin cells cannot be operated above 2.2 V (*vs.* Mg RE). Significantly, the potential stability reaches up to 3 V on Al and Ni (see Fig. 7b). This preliminary evidence hints that the halogen free anion in the electrolyte lessened the corrosive nature of the electrolyte. In our boron based electrolyte system, common, inexpensive materials of Al and Ni may serve as current collectors that enable over 2.5 V rechargeable magnesium battery systems.

Electrode–electrolyte compatibility is essential for developing practical rechargeable Mg batteries. In order to prove the compatibility of the $Mes_3B–(PhMgCl)_2$ boron based electrolyte solution with the magnesium ion intercalation cathode, the Mo_6S_8 Chevrel phase cathode, which can intercalate reversibly with Mg ions,[28] was synthesized by the molten salt method according to the literature.[14] The 2016-type coin cell comprising the boron based electrolyte, Mo_6S_8 Chevrel phase cathode, and Mg disc anode was charged and discharged at room temperature. Fig. 8a presents a typical potential–time curve with the discharge and charge voltage limits of 0.5 V and 1.7 V. The battery demonstrates reversible cycling behavior at a 0.1 C rate. Fig. 8b shows the capacity–voltage profiles at the 3rd, 20th and 50th cycles. The discharge capacity is around 80 mA h g^{-1} calculated based on the weight of the cathodes active mass. Two voltage plateaus at 1.2 V and 1.1 V are in accordance with Aurbach's results.[5] These experimental results confirm that the boron based electrolytes have good compatibility with the Mo_6S_8 Chevrel phase cathode. However, the low average operating voltage (1.1 V *vs.* Mg) of the Mo_6S_8 cathode cannot exert the advantage of the wide electrochemical window for the boron based electrolyte. In order to develop high-energy rechargeable Mg batteries using this boron based electrolyte, research towards finding suitable high potential cathode materials is in progress.

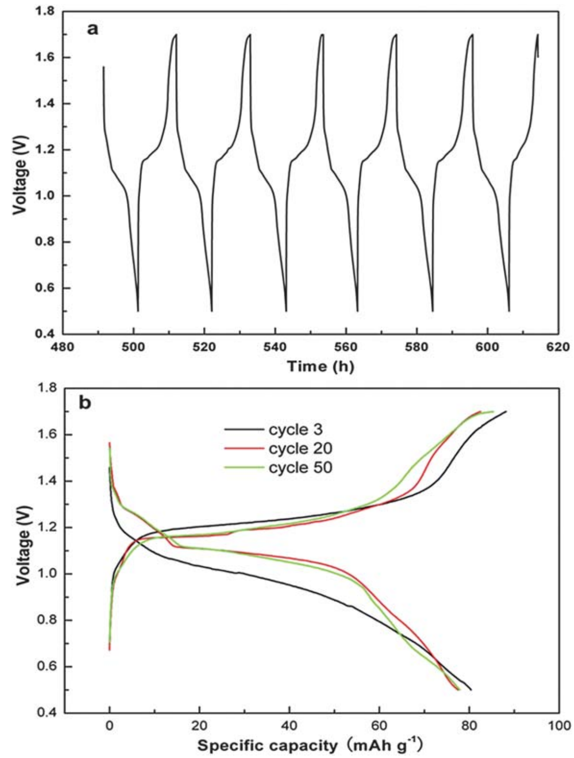


Fig. 8 (a) Typical voltage–time curves and (b) voltage–capacity profiles of rechargeable Mg batteries with $Mes_3B–(PhMgCl)_2$ boron based electrolyte solution, Mo_6S_8 cathode, and pure Mg metal as the anode.

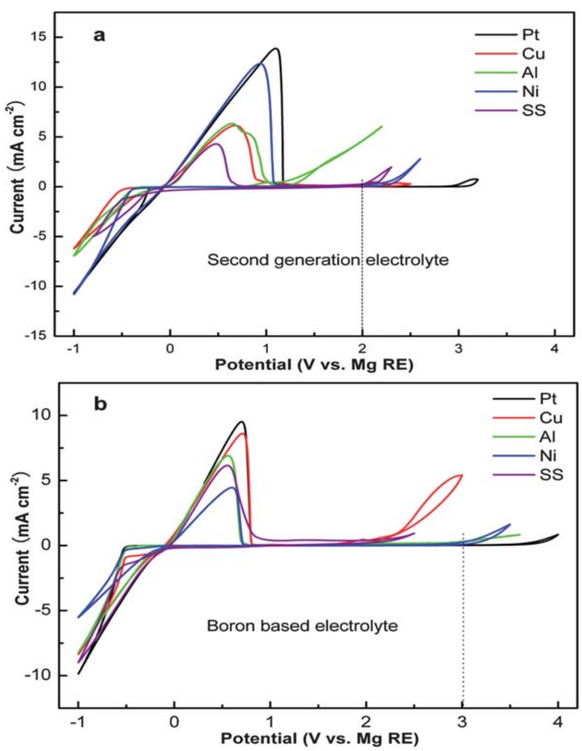


Fig. 7 Cyclic voltammograms of two electrolyte systems on several working electrodes: (a) $AlCl_3–(PhMgCl)_2/THF$, (b) $Mes_3B–(PhMgCl)_2/THF$.

4. Conclusions

We have identified the major equilibrium species in the novel boron based electrolyte solutions through characterizations by NMR, Raman spectroscopy, fluorescence spectroscopy, and crystallography analyses. The solution mainly contains the species of $Mg_2Cl_3^+$, $MgCl^+$, Ph_2Mg and the tetracoordinated boron anion $[Mes_3BPh]^-$. The high anodic stability of the boron based electrolyte is dominated by the intermediate product formed through weak interactions between the $[Mes_3BPh]^-$ anion and Ph_2Mg. Unfortunately, the exact intermediate product and function mechanism are still unclear due to the complicated ion compositions, interaction and solvation effect in the $Mes_3B–(PhMgCl)_2$ electrolyte solution. It is found that this electrolyte system possesses a powerfully recovered ability after partially losing its performance when exposed to air, which provides a simple way to overcome the potential shortcomings of electrolyte systems sensitive to air/moisture. Moreover, the less corrosive nature of the boron based electrolyte makes common and inexpensive materials of Al and Ni useful as current collectors for 2.5 V rechargeable magnesium battery systems. The good charge–discharge cycling performance of the Mg–Mo_6S_8 battery confirms that the boron based electrolyte could be practically used in rechargeable Mg battery systems. Further work will aim to find a suitable high potential cathode material for constructing high-energy rechargeable Mg batteries.

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